AnalystLab Africa - Week 3: Healthcare Operations & Financial Intelligence Dashboard

Task 1 Step 4: Star Schema Data Model Architecture

The data architecture is implemented using a Star/Snowflake hybrid design optimized for Power BI reporting. The configuration cleanly separates measurable operational events from descriptive categorical dimensions, ensuring efficient query performance and streamlined DAX aggregations.

Schema Design & Entity Relationships

Table Name	Classification	Source File	Record Count	Core Function & Primary Attributes
encounters	Core Fact	encounters.csv	27,891	Operational encounters containing encounter timestamps (`START`, `STOP`), financial values (`TOTAL_CLAIM_COST`, `PAYER_COVERAGE`, `BASE_ENCOUNTER_COST`), and relational keys.
procedures	Secondary Fact	procedures.csv	47,701	Patient clinical and procedural intervention records linked directly to corresponding encounter events.
patients	Dimension	patients.csv	974	Patient master demographic records, geographic locations, and baseline health details.
payers	Dimension	payers.csv	10	Insurance provider metadata, coverage parameters, and the synthetic self-pay/uninsured record.
organizations	Dimension	organizations.csv	N/A	Healthcare delivery networks, facilities, and organizational entity structures.

Configured Relational Connections (1-to-Many, Single Direction)

The relationships are enforced with single-directional cross-filtering from dimensions down to facts to maintain unambiguous filter propagation:

1. **patients[Id] (1) $\rightarrow$ encounters[PATIENT] (*)**
   Propagates patient demographic filters (gender, age cohorts, location) across clinical encounters.
2. **payers[Id] (1) $\rightarrow$ encounters[PAYER] (*)**
   Enables filtering and segmentation of financial totals by insurance provider and coverage category.
3. **organizations[Id] (1) $\rightarrow$ encounters[ORGANIZATION] (*)**
   Allows performance attribution and throughput tracking across healthcare provider entities and clinics.
4. **encounters[Id] (1) $\rightarrow$ procedures[ENCOUNTER] (*)**
   Enables deep-dive analysis linking specific procedural codes and execution costs back to their parent encounter visit.

Task 1 Step 5: Documented Data Transformations & Technical Justifications

Step #	Target Field / Table	Applied ETL / Modeling Transformation	Technical Justification
1	`START, STOP, BIRTHDATE (All Tables)`	**Datetime Type Parsing:** Cast raw string timestamps into native Power BI `Date/Time` and `Date` data types.	Required to support chronological sorting, accurate duration calculations, and DAX time-intelligence functions (e.g., Year-over-Year, Year-to-Date, and monthly trends).
2	`REASONDESCRIPTION (encounters, procedures)`	**Missing Description Handling:** Replaced `null` values with `"Routine / Wellness / Unspecified"`.	Over 70% of encounter records correspond to routine preventive care without diagnostic trigger codes. Eliminates blank categories in visuals and prevents broken slicer filters.
3	`encounters Table`	**Duration Calculation:** Added calculated column: `Duration_Hours = DATEDIFF(encounters[START], encounters[STOP], HOUR)`	Provides an operational metric to distinguish brief outpatient/urgent care visits from extended inpatient admissions, measuring bed utilization and throughput.
4	`encounters Table`	**Patient Liability Calculation:** Added column/measure: `Patient_Liability = encounters[TOTAL_CLAIM_COST] - encounters[PAYER_COVERAGE]`	Separates costs covered by third-party payers from out-of-pocket liabilities incurred directly by patients.
5	`payers Table`	**Uninsured Entity Resolution:** Validated and retained the synthetic `NO_INSURANCE` entity mapping.	Accurately segments self-pay encounters (representing 48.71% of dataset encounters) while preserving referential integrity across the data model without record loss.